

Lecture 10: Forgetting

PSYCH/COGS 140M 2026.05.12



"Mr. Osborne, may I be excused? My brain is full."

Notes

Unit 3: *Internal influences on memory*

Lecture 9: *Cognitive Control* - “top-down” influences on learning and memory.

Lecture 10 (today): *Forgetting* - How, why, when

Lecture 11: *Distortion and False memory*

Lecture 12: *Emotional memories*

Quiz #3

Forgetting

What is Forgetting and
Why do we do it?

Is forgetting a modern disease?

WELLNESS

Why Forgetting Is Good For Your Memory

Why Do We Remember Certain Things, But Forget Others?

Why Do We Forget Things?

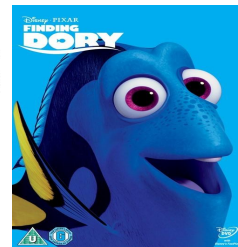
Memory loss identified years before Alzheimer's symptoms appear

February 6, 2018 | University College London

Could The Best Memory System Be One That Forgets?

Forgetting Things Could Actually Be Making You

What Finding Dory Teaches Us About Memory Loss



What is forgetting?



When we can't find a memory?

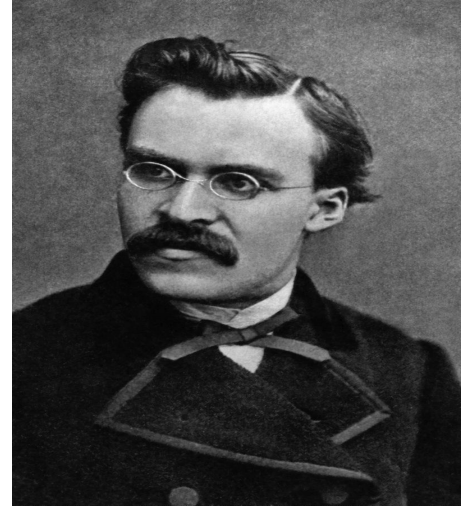


When memories fade away?

What is forgetting?

“The existence of forgetting has never been proved: We only know that some things don't come to mind when we want them.”

— Friedrich Nietzsche



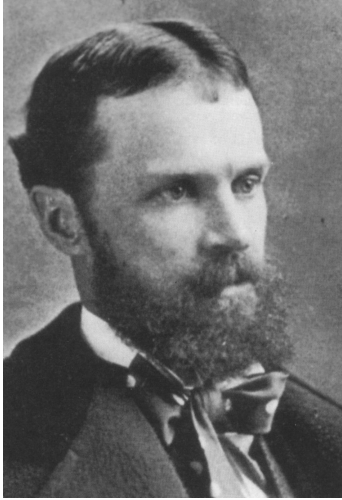
1844 - 1900

Forgetting: good or bad?

Forgetting makes us efficient – we only hold onto what we need.

Forgetting is just a limitation — it's when memory fails.

Benefits of forgetting



William James

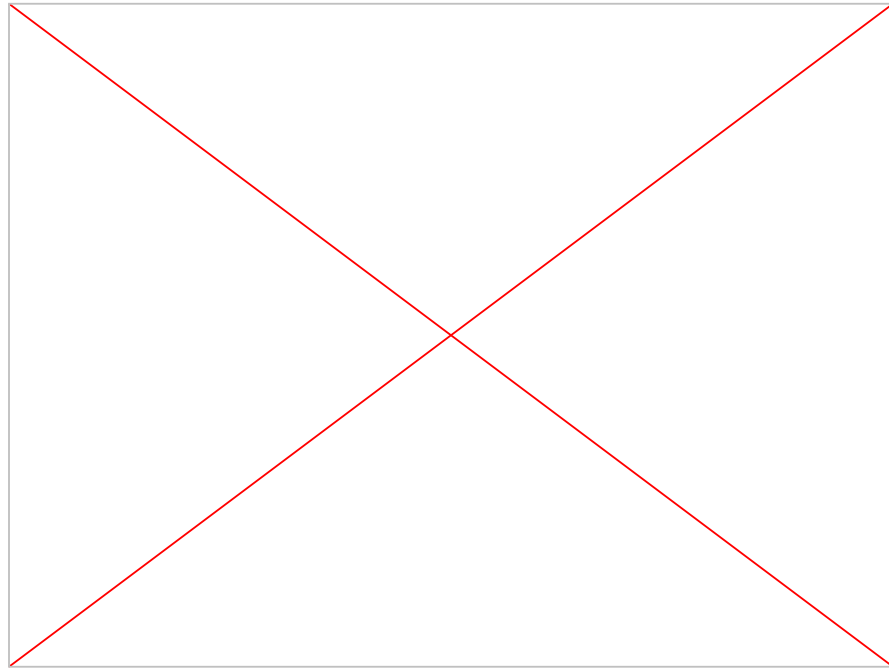
Principles of Psychology

"**Selection** is the very keel on which our mental ship is built. And in the case of memory its utility is obvious. If we remembered everything, we should on most occasions be as ill off as if we remembered nothing."

What if we didn't forget?

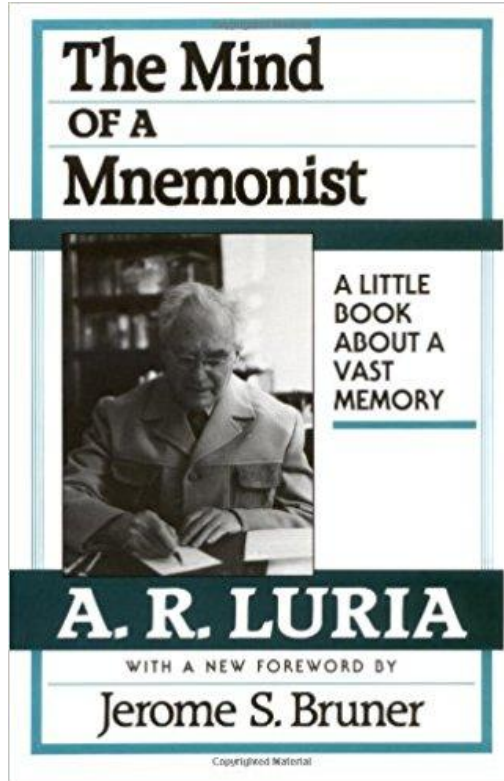
Examples of superior autobiographical memory

- Clip from “Endless Memory” - 60 Minutes, CBS News



What if we didn't forget?

“S.”, a mnemonist studied by Luria, was a man who almost *never* forgot.



What if we didn't forget?

“S.”, a mnemonist studied by Luria, was a man who almost *never* forgot. ***Is there a cost to remembering so much?***

Upon being read a passage at a rapid pace:

“No . . . This is too much. Each word calls up images; they collide with one another and the result is chaos. I can't make anything out of this.”

Describing reading a book:

“The things I see when I read aren't real . . . but all the same my images led me there automatically . . . This makes for a tremendous amount of conflict . . . What happens is I can't read, can't study, for it takes up such an enormous amount of my time.”

Outline

- Some thoughts on Forgetting
- Why do we forget?

Why do we forget?

- Failure to encode
- Decay over time
- Ineffective retrieval cues
- Interference
- Remembering
- We try *not* to remember (*directed forgetting*)

Why do we forget?

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Failure to encode

- Frequently, when we say that we forgot something, it was never well learned in the first place
- Think of all the factors affecting encoding that we covered...
- E.g., divided attention
 - You're distracted when you walk in the door, and you don't attend to where you place your keys
 - Later you "forget" where you put them

Why do we forget?

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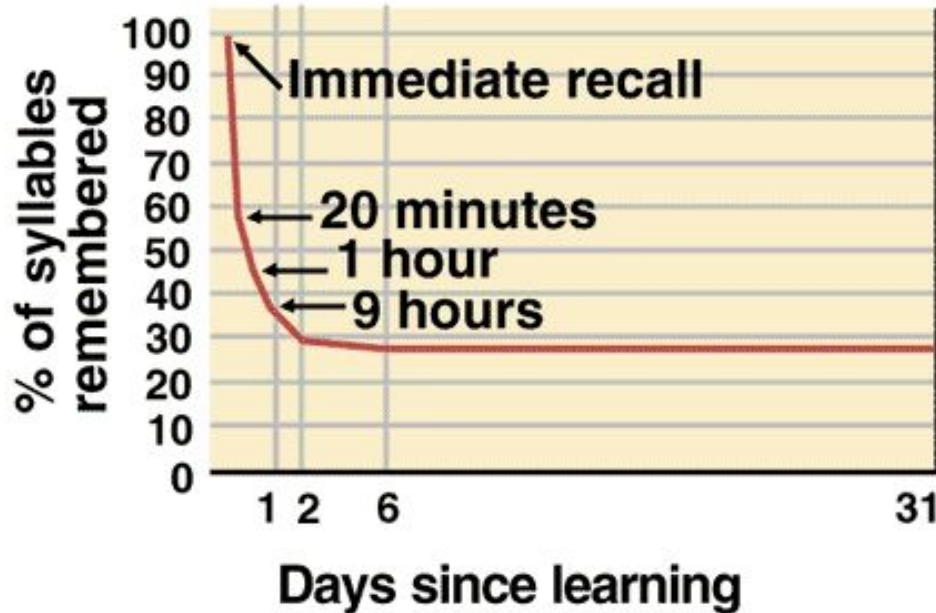
Decay



(S. Dali, 1931, *Persistence of Memory*)

Decay

Ebbinghaus's Forgetting Curve

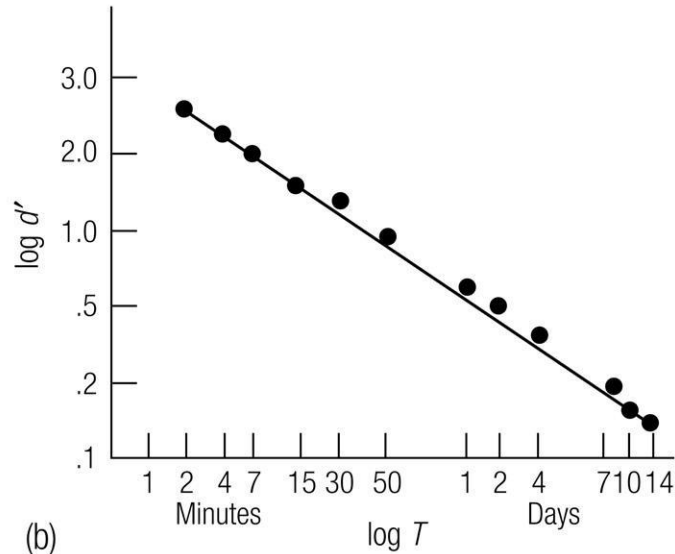
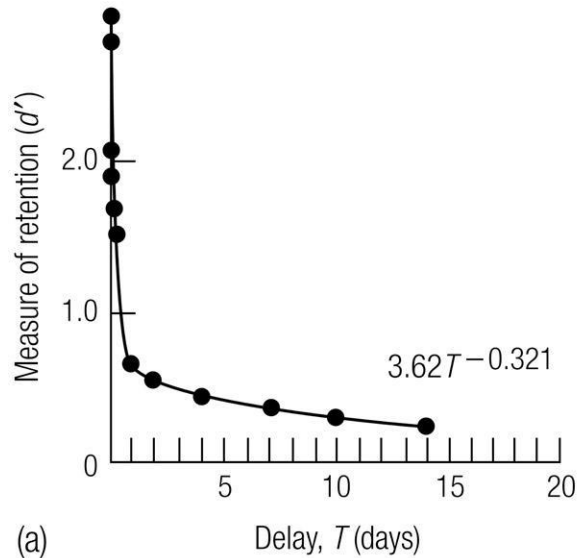


- the forgetting function is *negatively accelerated*
- more and more delays yield smaller and smaller losses

Decay

Wickelgren (1975) word recognition

Negatively accelerated forgetting curve, like Ebbinghaus' function



The forgetting curve in real life

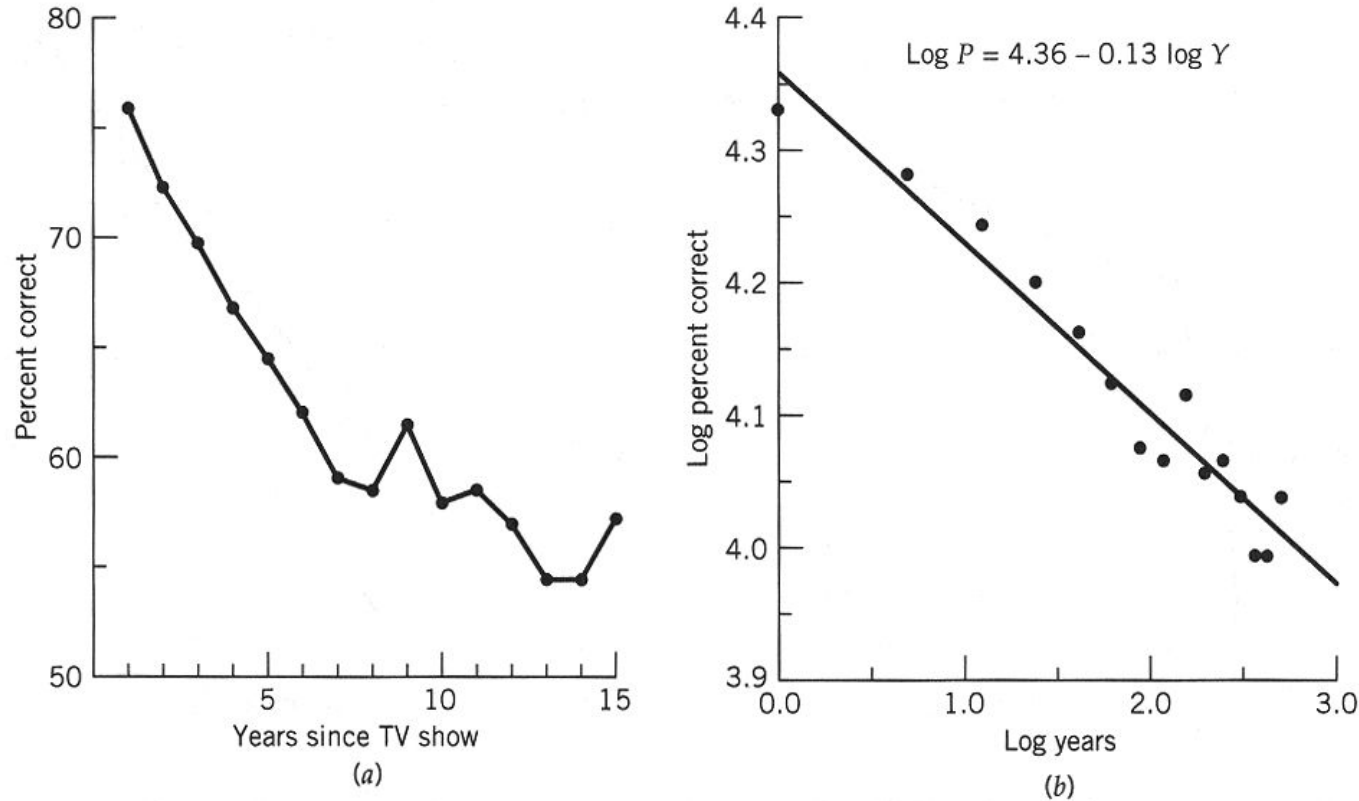


FIGURE 7.4 (a) Probability of recognizing a TV show as a function of time since cancellation; (b) the same data on a log-log scale. (From Squire, 1989.)

Decay over time

- As time passes, memory is worse
- Trace decay theory
 - Memory traces (i.e., connections between neurons) weaken or degrade over time
- Theory is difficult to conclusively test!
 - Do connections simply grow weaker?
 - Or do other (confounding) factors explain weakening over time?
 - What might these factors be?

Why do we forget?

- Failure to encode
- Decay over time
- **Ineffective retrieval cues**
- Interference
- Remembering
- We try *not* to remember (*directed forgetting*)

Ineffective retrieval cues

username

AARON

password

??BOTH OF THEM??

PASSWORD HINT

**It's the name
of your
favorite cat.**

Memory Tests Vary in Cue Strength

Demo: Study the following list of words!

Memory Tests Vary in Cue Strength

Elm

Iron

Apple

Shirt

Palm

Fork

Banana

Chair

Pants

Oak

Pear

Hat

Memory Tests Vary in Cue Strength

Free recall:

Fruits:

B_____

P_____

A_____

Clothing:

S_____

H_____

P_____

Trees:

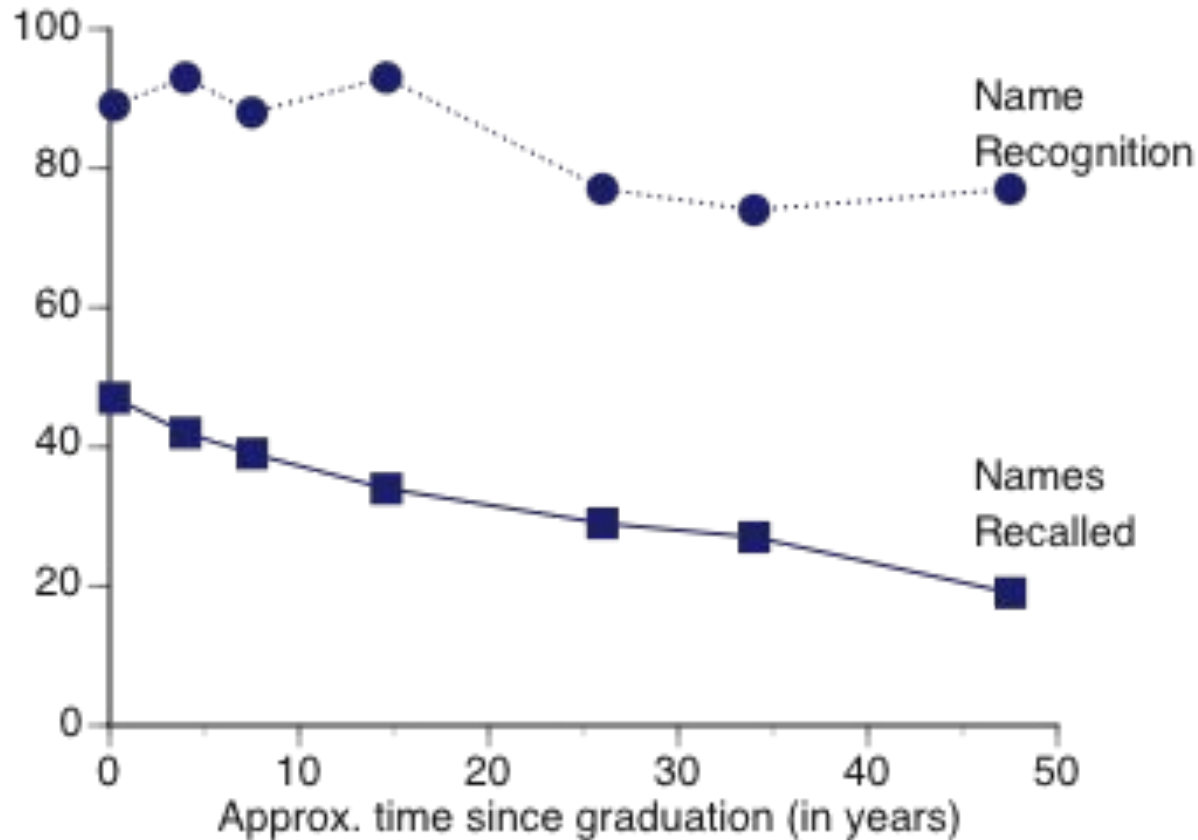
O_____

P_____

E_____

Remembering high school classmates

Percent Correct



(Bahrick et al., 1975)

Role of Retrieval Cues

The amount of “forgetting” that is observed will depend on the specific cues that are used to test memory!

Why do we forget?

- Failure to encode
- Decay over time
- Ineffective retrieval cues
- **Interference**
- Remembering
- We try *not* to remember (*directed forgetting*)

What is interference?

- The storage of similar *memory traces* interferes with our ability to retrieve a target memory
- When a *retrieval cue* is associated with multiple memories, they **compete** with one another

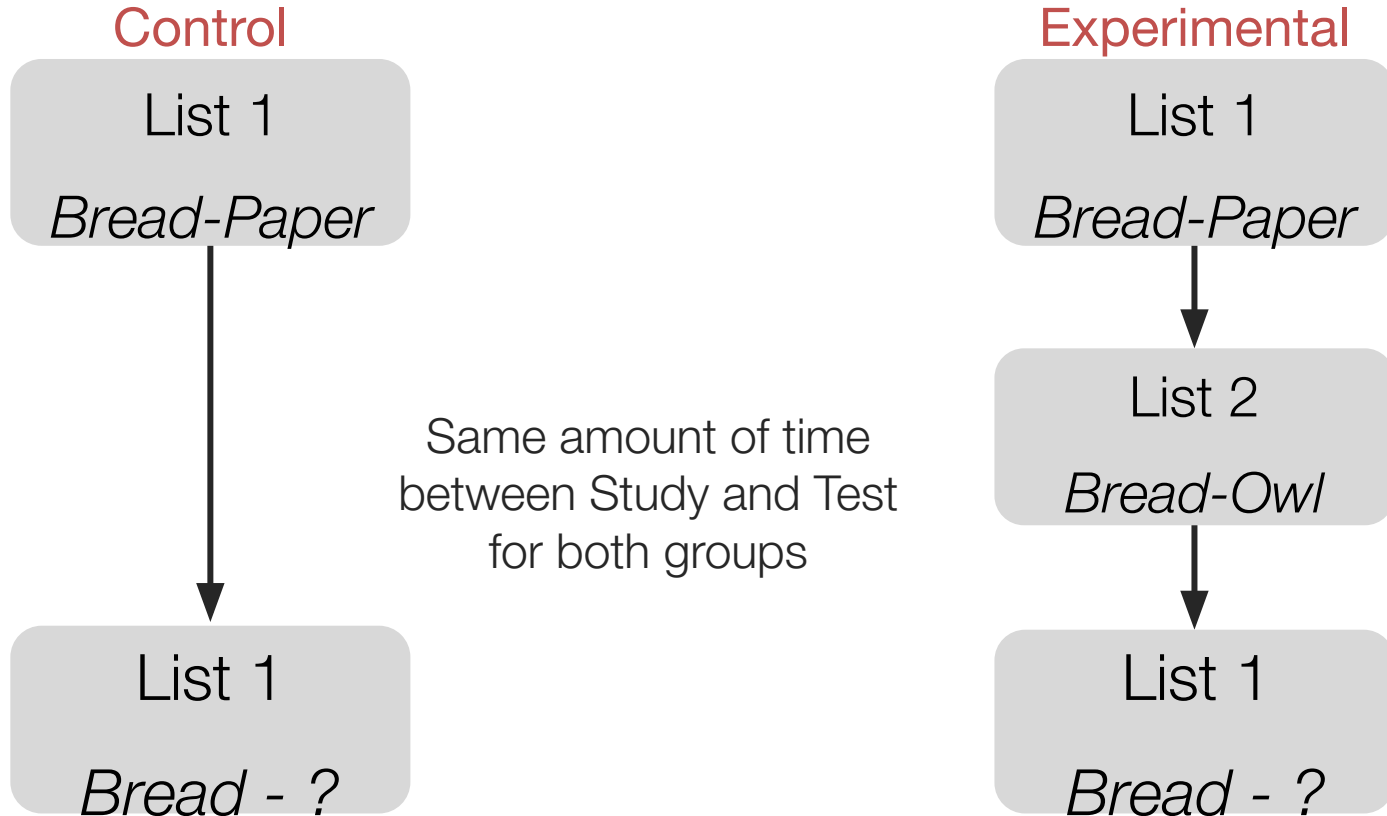
Interference based on order of learning

- **Retroactive interference**
 - **Newer information** interferes with the ability to retrieve **older** information
- **Proactive interference**
 - **Older information** interferes with the ability to retrieve **newer** information

Retroactive interference in daily life

- Forgetting your old address after moving to a new one
- Forgetting what you had for dinner last week (because you had several dinners since then)
- Forgetting information from the first midterm after studying for the second midterm

Retroactive interference

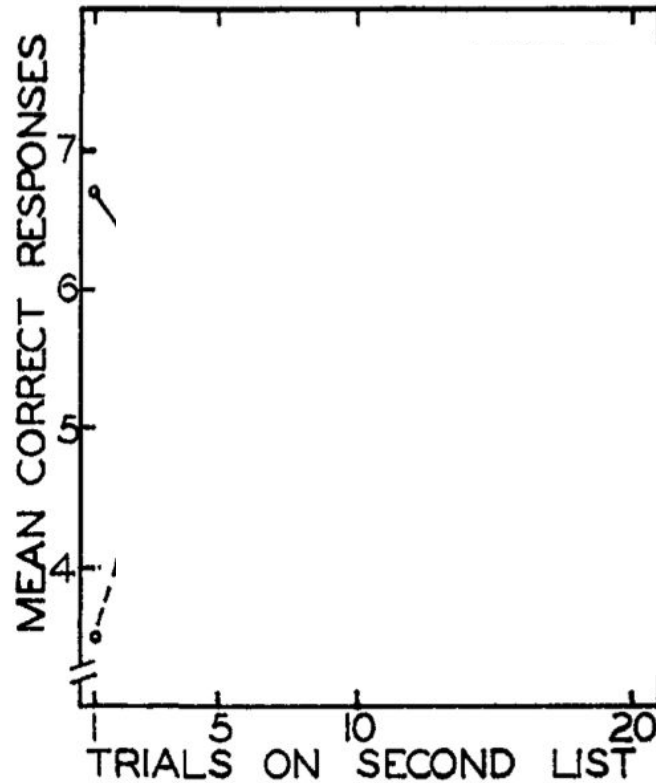


Retroactive interference

Typical results of such experimental designs show the following:

- A similar second list impairs the ability to recall items from the first list (relative to control)
- Increased training on the second list further impairs memory for the first list

Barnes & Underwood (1959)



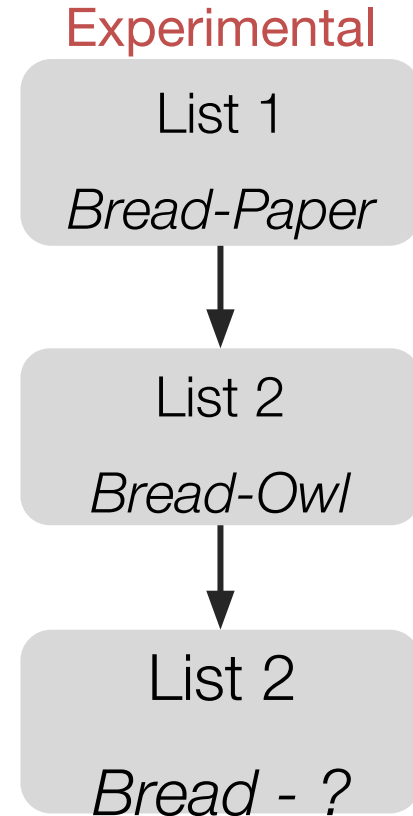
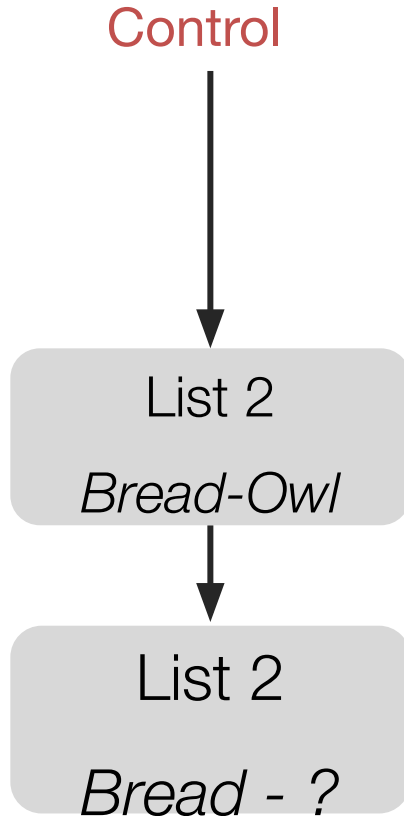
Recall List 2
Bread-Owl

Recall List 1
Bread-Paper

Proactive interference in daily life

- In January, writing the previous year's date instead of the new date
- Walking to yesterday's parking spot instead of today's spot
- Calling your friend by another's name
 - More common with more similar names

Proactive interference

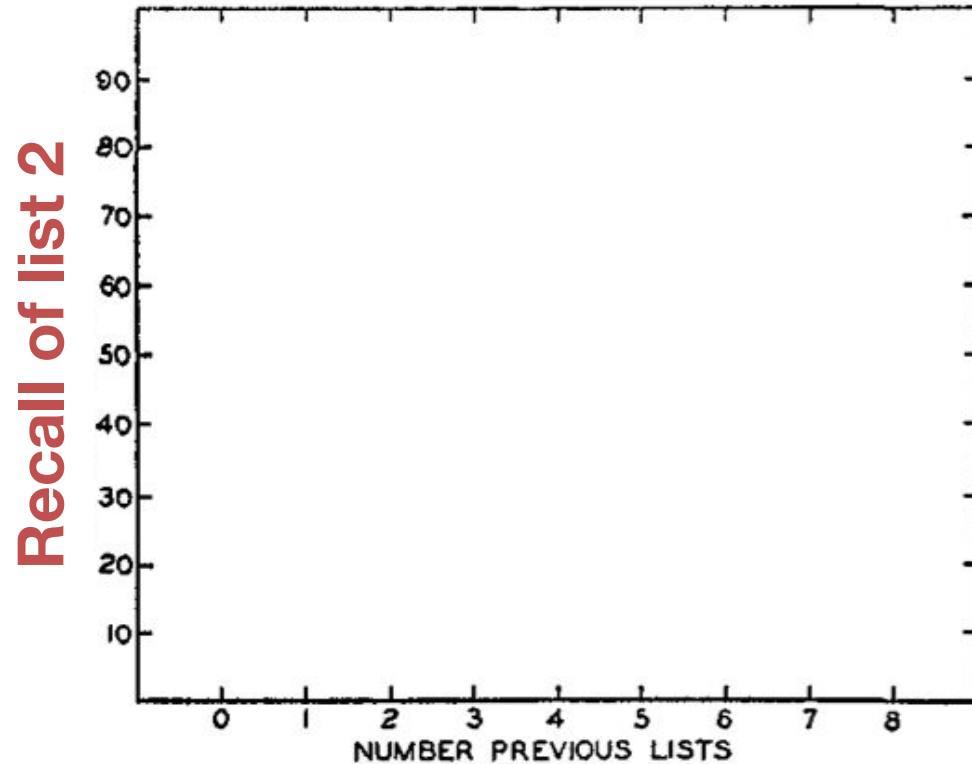


Proactive interference

Typical results of such experimental designs show that:

- Recall for list 2 is impaired when similar material is first presented in list 1 (relative to control)
- The greater the amount of previously learned material, the worse memory is for newer material

Underwood (1957)



Interference: Cue Overload

- When a single retrieval cue has many *associates*, the potential for interference is higher

Interference: Cue Overload

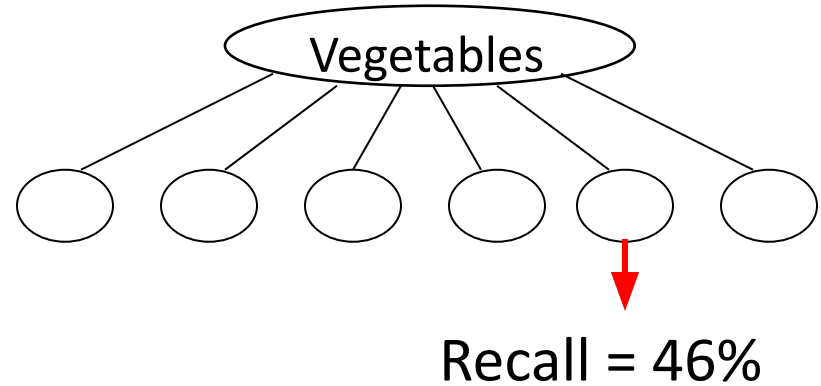
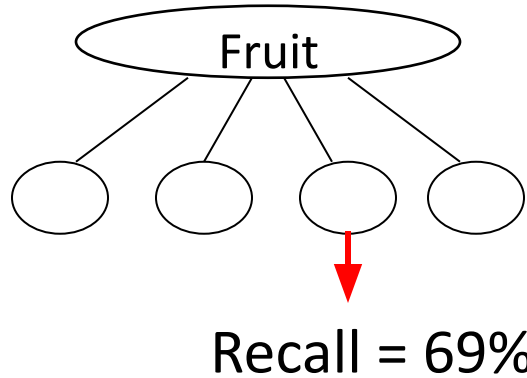
Bill's Grocery List

Milk	Napkins
Eggs	Chips
Bagels	Water
Coke	Steak
Bread	Sugar
Rice	Flour
Cheese	Butter
OJ	Corn

Steve's Grocery List

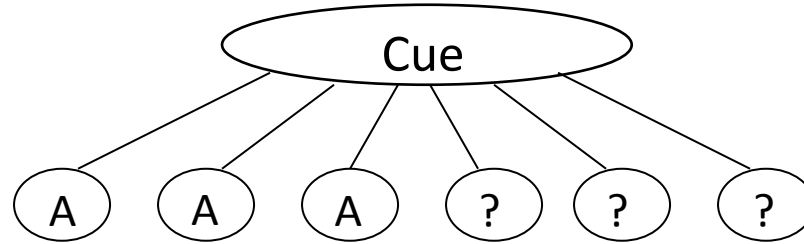
Milk
Bread
Cheese
OJ

Interference: Cue Overload



Better recall when there are fewer associates

Interference: Part-set cuing



Providing some of the associates of a cue can
'block' access to the remaining associates

Example: Giving someone a list of 25 of the states
actually makes it *harder* to recall the remaining 25!

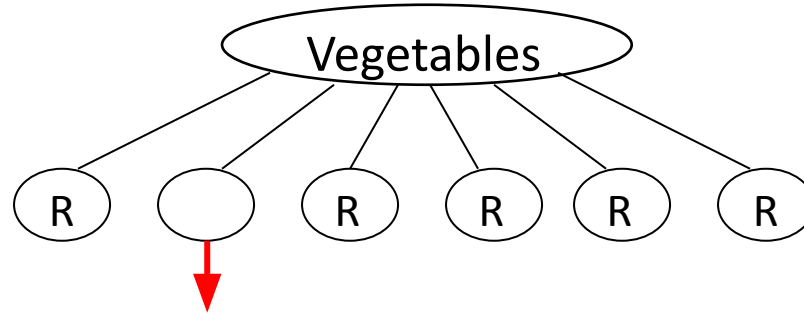
Summary: Interference and forgetting

- We store each day's events in long-term memory
- On subsequent days, we do many similar things (routines, habits)
- These similar events add interfering traces into memory
- Eventually, interference impairs our ability to remember the past → forgetting

Why do we forget?

- Failure to encode
- Decay over time
- Ineffective retrieval cues
- Interference
- Remembering
- We try *not* to remember (*directed forgetting*)

Output Interference



The successfully recalled
associates *interfere* with recall
of the remaining associate

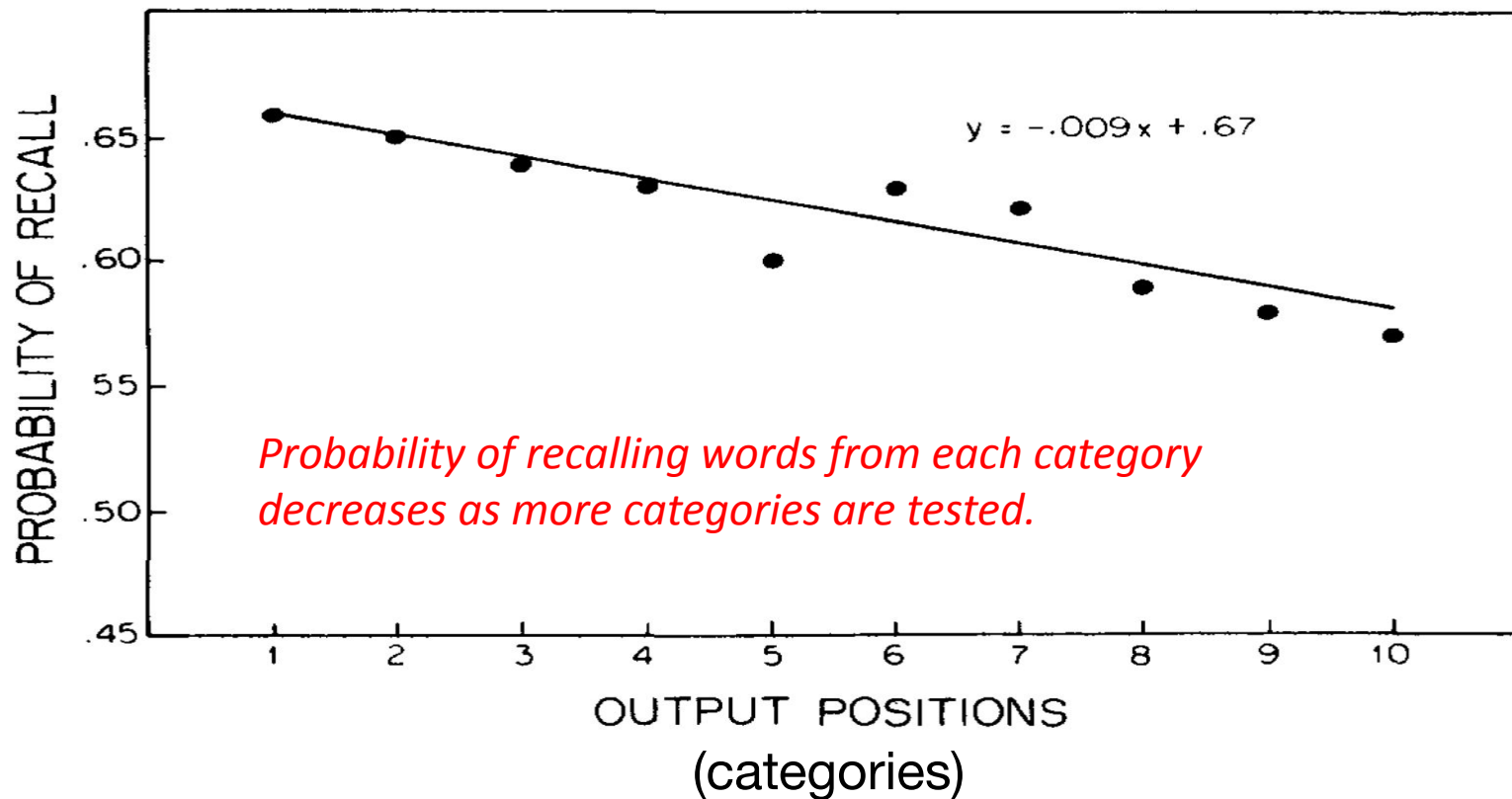
- Output interference occurs as a result of having just retrieved something else

Output Interference

Experiment by Roediger & Schmidt, 1980

- Subjects studied lists of words from various categories (e.g., fruits, vegetables, trees, etc.)
- Memory was then tested for one category at a time
- Question: does memory recall decrease as more categories are tested (i.e., as more items are recalled)?

Output Interference



Retrieval-Induced Forgetting

Idea that retrieving some memories can result in long-lasting weakening (forgetting) of related memories

Retrieval-Induced Forgetting

STUDY

FRUITS:

Fruit - Apple
Fruit - Banana
Fruit - Orange

TOOLS:

Tool - Hammer
Tool - Saw
Tool - Wrench

RETRIEVAL PRACTICE

FRUITS:

Fruit - A_
Fruit - O_

TOOLS:

TEST

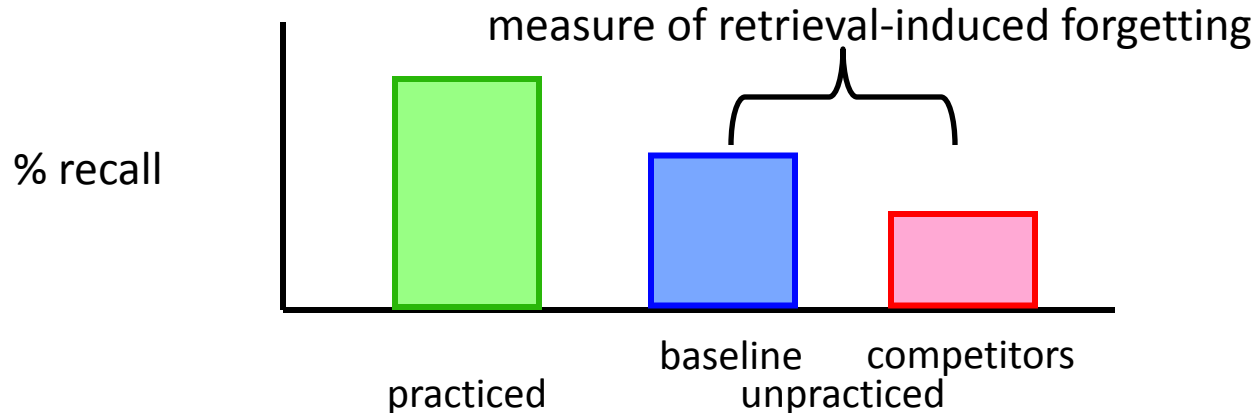
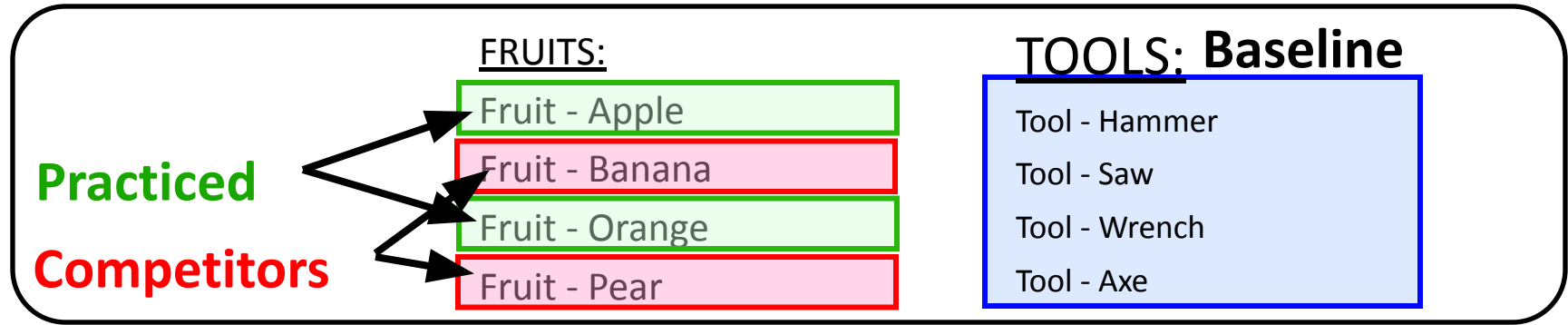
FRUITS:

Fruit - A_
Fruit - B_
Fruit - O_
Fruit - P_

TOOLS:

Tool - H_
Tool - S_
Tool - W_
Tool - A_

Retrieval-Induced Forgetting



Retrieval-Induced Forgetting

Can retrieval-induced forgetting be ‘socially transmitted?’

Experiment by Cuc, Koppel, and Hirst (2007)

- Two people read same story
- One person (speaker) then talks about *part* of the story

Listener



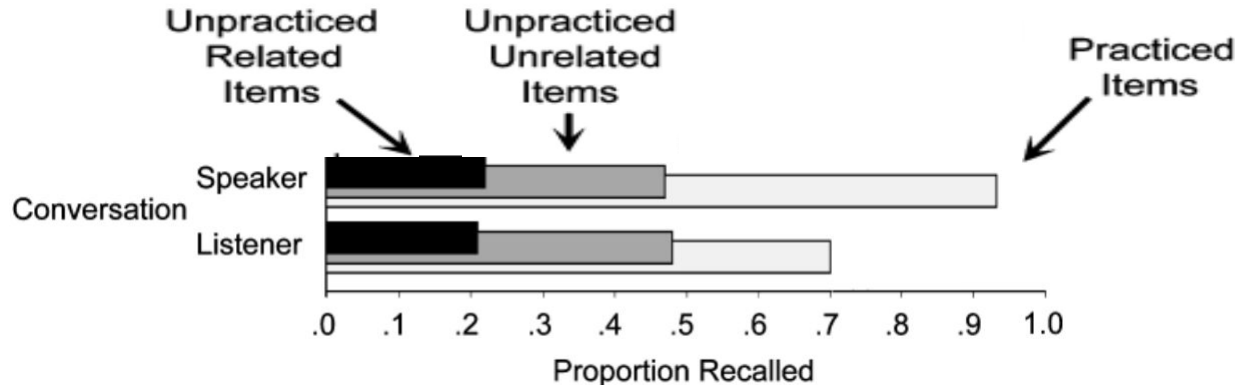
Speaker

Retrieval-Induced Forgetting

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Why do we forget?

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- *We try not to remember (directed forgetting)*

Think / No-Think Task

STUDY

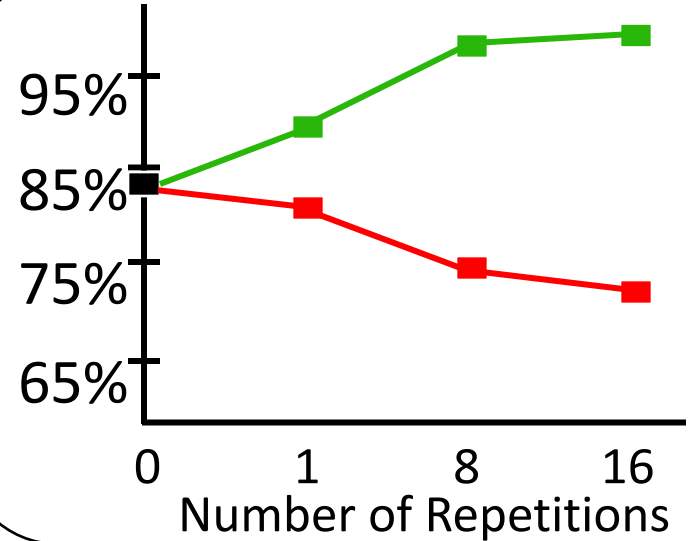
Ordeal - Roach
Journey - Pants
Flag - Sword

THINK / NO-THINK

Flag -
Ordeal -

TEST

Ordeal - R_
Journey - P_
Flag - S_



More practice 'not remembering' (No Think trials) leads to worse memory when items are later tested.

Why does “not thinking” lead to forgetting?

- Option 1: **Blocking**
 - On ‘no think’ trials, subjects think of something else (form new memories)
 - Individual memories don’t get weakened
 - New memories just get in the way (similar to retroactive interference)
- Option 2: **Suppression**
 - Individual memories *do* get weakened

Think / No-Think Task

STUDY

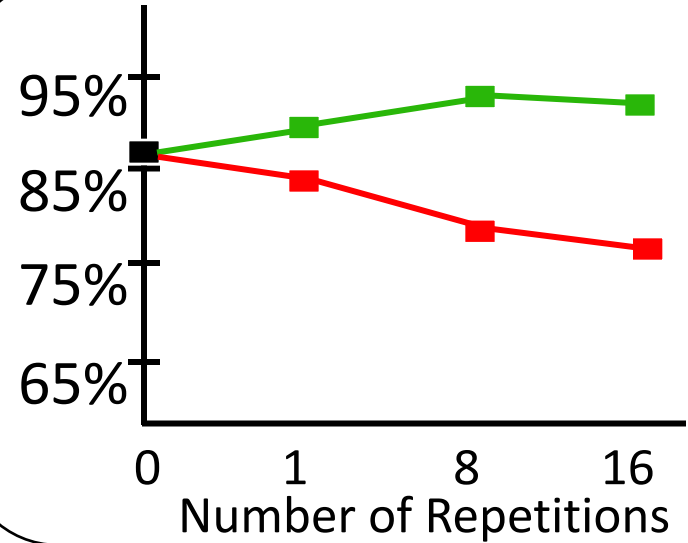
Ordeal - Roach
Journey - Pants
Flag - Sword

THINK / NO-THINK

Flag -
Ordeal -

TEST

Insect - R_
Clothes - P_
Weapon - S_



Forgetting of No-Think items evident even with a new cue:
Cue-independent forgetting

Neural Mechanisms of Forgetting?

One potential cause: neurogenesis in the hippocampus

- As in humans, mice show periods of **infantile amnesia**
- Infantile amnesia tends to coincide with **neurogenesis** (the creation of new neurons) in the hippocampus
 - neurogenesis is most prevalent in early development
 - rare in adults (besides **dentate gyrus** and **olfactory bulb**)



Neural Mechanisms of Forgetting?

Akers et al. 2014 tested whether forgetting of older memories can be influenced by *manipulating* neurogenesis

- They found that:
 - *Increasing* neurogenesis led to *greater* forgetting of older memories
 - *Decreasing* neurogenesis led to *reduced* forgetting of older memories
- Thus, neurogenesis in the hippocampus disrupts memory circuits & plays a role in forgetting

Next Up:

- False Memories